

Rajiv Gandhi University of Health Sciences, Karnataka
Sixth Semester B. Pharm Degree Examination – 14-Dec-2020

Time: Three Hours

Max. Marks: 75 Marks

PHARMACEUTICAL BIOTECHNOLOGY

Q.P. CODE: 5027

Your answers should be specific to the questions asked.
Draw neat labeled diagrams wherever necessary.

LONG ESSAYS (Answer any Two)

2 x 10 = 20 Marks

1. What are biosensors? Write its principle and its functions.
2. Write in detail the production of Insulin by genetic engineering method.
3. Write in detail the different types of fermenters and its applications.

SHORT ESSAYS (Answer any Seven)

7 x 5 = 35 Marks

4. How to make sodium alginate beads in calcium chloride solution?
5. What is PCR? Write the working principle of PCR.
6. Explain the structure of immunoglobulins.
7. Write the production of killed bacterial vaccines.
8. Write the production of monoclonal antibodies.
9. Write in brief the western blotting technique and mention its application.
10. Write in brief about transformation with suitable examples.
11. Write briefly about different types of mutation and give its significance.
12. Write in brief the Collection, Processing and Storage of whole human blood.

SHORT ANSWERS (Answer All)

10 x 2 = 20 Marks

13. Name any four natural polymers used for immobilization.
14. Name any four disadvantages of immobilization.
15. Write any four applications of interferons.
16. Define toxoids. Give two examples.
17. Name any two blood products with applications.
18. What is hypersensitivity? Write the types.
19. What is microbial biotransformation? Give two examples.
20. What is downstream processing? Give two examples.
21. Expand ELISA and write two applications.
22. Applications of plasma substitutes.

Rajiv Gandhi University of Health Sciences, Karnataka
Sixth Semester B. Pharm Degree Examination - 10-Dec-2020

Time: Three Hours

Max. Marks: 75 Marks

BIOPHARMACEUTICS AND PHARMACOKINETICS

Q.P. CODE: 5026

Your answers should be specific to the questions asked.
Draw neat labeled diagrams wherever necessary.

LONG ESSAYS (Answer any Two)

2 x 10 = 20 Marks

1. List the various processes through which drugs can cross the biological membrane. Describe absorption of drugs from non per oral extra-vascular routes.
2. Explain various methods to enhance the dissolution rate of poorly soluble drugs.
3. Explain determination of pharmacokinetic parameters from plasma concentration data after administration of drug I.V.bolus.

SHORT ESSAYS (Answer any Seven)

7 x 5 = 35 Marks

4. Define biopharmaceutics and discuss its role in formulation development.
5. Write in detail about protein binding and its significance.
6. Write a note on renal excretion of drugs.
7. Explain bioequivalence studies.
8. Discuss about the blood level curves of a drug administered by I.V. infusion and oral routes.
9. What are pharmacokinetic models? Explain various types with their significance.
10. Estimate one compartment model parameters by using the method of residuals.
11. Explain about Michaelis - Menten's equation?
12. Write a note on determination of K_m and V_{max} at steady state concentration.

SHORT ANSWERS (Answer All)

10 x 2 = 20 Marks

13. What is hepatic first pass effect?
14. What is the influence of GI pH on drug absorption?
15. Enlist objectives of bioavailability studies.
16. Define clearance. What is its unit?
17. Define C_{max} and AUC.
18. Define apparent volume of distribution and give the mathematical equation to calculate it.
19. Define loading dose and maintenance dose.
20. What do you mean by central and peripheral compartment in two compartment model?
21. Define dose dependent kinetics.
22. Compare the concept of linear and non linear pharmacokinetics.

Rajiv Gandhi University of Health Sciences, Karnataka
Sixth Semester B. Pharm Degree Examination - 08-Dec-2020

Time: Three Hours

Max. Marks: 75 Marks

HERBAL DRUG TECHNOLOGY

Q.P. CODE: 5025

Your answers should be specific to the questions asked.
Draw neat labeled diagrams wherever necessary.

LONG ESSAYS (Answer any Two)

2 x 10 = 20 Marks

1. Explain the importance of garbling, drying and preservation in the processing of herbal raw materials.
2. Describe the herbs used in hair care in detail with examples.
3. Write a detailed note on patenting of traditional knowledge and natural products with examples.

SHORT ESSAYS (Answer any Seven)

7 x 5 = 35 Marks

4. Explain any two methods for the determination of alcohol content in asavas.
5. Explain the health benefits and role of nutraceuticals in diabetes.
6. Explain the health benefits and role of ashwagandha and ginseng as nutraceuticals.
7. Describe the possible side effects and interactions of garlic and ephedra.
8. Explain the diluents and viscosity builders from natural source with examples.
9. Write a note on fixed oils and waxes used in herbal cosmetics with examples.
10. Write the significance of quality control of raw materials and extracts, explain.
11. Explain the basic principles of Siddha and Unani.
12. Give a brief account on plant-based industries and institutions in India.

SHORT ANSWERS (Answer All)

10 x 2 = 20 Marks

13. Write any two plant-based bioinsecticides and their biological source.
14. What are traditional medicines? Name any two Ayurvedic formulations.
15. What are the source, active constituents and uses of honey?
16. What are the source, active constituents and uses of ginkgo biloba?
17. Name any two natural sweeteners and give their biological source.
18. Mention any two natural flavoring agents and give their biological source.
19. Give two evaluation parameters for herbal skin care products.
20. What is bioprospecting and biopiracy?
21. Define breeder's right?
22. What is schedule T?

Time: Three Hours

PHARMACEUTICAL QUALITY ASSURANCE
Q.P. CODE: 5028

Your answers should be specific to the questions asked.
Draw neat labeled diagrams wherever necessary.

LONG ESSAYS (Answer any Two)

2 x 10 = 20 Marks

1. Explain ICH Q1 guidelines for stability testing of drug and drug product.
2. Discuss briefly cGMP guidelines for construction, maintenance and sanitation of pharmaceutical unit.
3. Explain the objectives and scope of GLP.

SHORT ESSAYS (Answer any Seven)

7 x 5 = 35 Marks

4. Write the elements of ISO 9000.
5. Write the procedure for NABL accreditation.
6. How is the cross contamination prevented in dispensing and production areas.
7. How do you audit vendor for ensuring purchase specification?
8. Explain the QC test for secondary packaging material.
9. Discuss the procedure for conducting non clinical laboratory studies.
10. Discuss the SOP for disposal of waste in pharmaceutical unit.
11. Elaborate on master formula record.
12. Write the procedure for qualification of UV- Visible spectrophotometer.

SHORT ANSWERS (Answer All)

10 x 2 = 20 Marks

13. Write any four elements of TQM.
14. Define QA and QC.
15. What is cross contamination and mix up contamination?
16. Name any four types of closures.
17. What is DMF? Give its importance?
18. What is quality audit and quality review?
19. Define calibration, qualification and validation.
20. Write briefly the scope of validation.
21. Name the parameters used for analytical method validation.
22. What is the difference between QMS and EMS?

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PHARMACOLOGY - III

Q.P. CODE: 5024

Your answers should be specific to the questions asked.
Draw neat labeled diagrams wherever necessary

2 x 10 = 20 Marks

LONG ESSAYS (Answer any Two)

1. Classify anti ulcer agents with examples. Write mechanism of action and therapeutic uses of PPIs.
2. Classify Penicillins. Write mechanism of action, adverse effects and uses of Penicillin- G.
3. Classify antitubercular agents. Explain mechanism of action of INH and Rifampicin.

7 x 5 = 35 Marks

SHORT ESSAYS (Answer any Seven)

4. What is bronchial asthma? Classify drugs used in its treatment.
5. Classify antibiotics on the basis of mechanism of action along with examples.
6. Write mechanism of action and uses of streptomycin.
7. Write mechanism of action, adverse effects and uses of Zidovudin.
8. Write a flow chart showing life cycle of plasmodium species.
9. Describe mechanism of action and therapeutic uses of alkylating agents.
10. Explain mechanism of action and uses of cyclosporine.
11. Outline the steps involved in the elimination of orally ingested poisons.
12. Explain the terms acute, subacute and chronic toxicity studies along with their purpose.

10 x 2 = 20 Marks

SHORT ANSWERS (Answer All)

13. Enlist non systemic antacids.
14. Write uses of expectorants and Antitussives.
15. What are nasal decongestants? Give examples.
16. Mention four classes of antibiotics acting by inhibiting cell wall synthesis.
17. Mention the drugs used for treatment of filariasis.
18. Mention common causative organisms for UTI.
19. Enlist corticosteroids.
20. What is teratogenicity? Give an example.
21. Mention four biological rhythms.
22. What is biological clock?

Time: Three Hours

Max. Marks: 75 Marks

MEDICINAL CHEMISTRY-III

Q.P. CODE: 5023

Your answers should be specific to the questions asked.

Draw neat labeled diagrams wherever necessary.

LONG ESSAYS (Answer any Two)

2 x 10 = 20 Marks

1. Write the structure and uses of three drugs each from cephalosporins and tetracyclines. Discuss in detail SAR of tetracyclines.
2. What are antimalarial agents? Classify them with examples. Give the mechanism of action of quinolones and outline the synthesis of Pamaquine.
3. What are therapeutic agents? Classify sulphonamides with examples. Outline the synthesis of Trimethoprim and Sulphamethoxazole.

SHORT ESSAYS (Answer any Seven)

7 x 5 = 35 Marks

4. Write a note on beta lactamase inhibitors.
5. What are aminoglycosides? Write the mechanism of action and chemistry of aminoglycosides antibiotics.
6. Write the SAR of 7-chloro 4-amino quinolines.
7. Write the chemistry and of Erythromycin.
8. What are first line anti-tubercular drugs? Write the structure of any two anti- tubercular drugs. Give the synthesis of para amino salicylic acid.
9. What are antiviral agents? Classify them with suitable examples.
10. Add a note on synthetic antifungal agents. Give the synthesis of Tolnaftate.
11. Define and classify anthelmintics. Write the synthesis of Diethyl carbamazine citrate.
12. Explain the modern concept of rational drug design.

SHORT ANSWERS (Answer All)

10 x 2 = 20 Marks

13. Write the structure and uses of Ampicillin.
14. Write the structure and uses of Chloramphenicol.
15. Give the synthesis of Isoniazid.
16. Outline the synthesis of Nitrofurantoin.
17. Give the structure and uses of any one antiviral drug.
18. Define anti-tubercular agents? Name the causative organism for tuberculosis.
19. List out antifungal agents having imidazole nucleus.
20. What are antiprotozoal agents? Give the structure and uses of Ornidazole.
21. Give the synthesis of Mebendazole.
22. Mention the electronic parameters used in QSAR.
